Abstract Algebra III: Assignment - Week 4

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Problem 1

Let $M, N, M_i (i \in I)$ and $N_j (j \in J)$ be R-modules. Prove the following isomorphisms of abelian groups.

(1) $\mathrm{Hom}_R\left(\bigoplus_{i\in I} M_i, N\right) \simeq \prod_{i\in I} \mathrm{Hom}_R(M_i, N)$;

(2) $\mathrm{Hom}_R\left(M, \prod_{j\in J} N_j\right) \simeq \prod_{j\in J} \mathrm{Hom}_R(M, N_j)$.

Solution: **(1)** Let $X = \mathrm{Hom}_R\left(\bigoplus_{i\in I} M_i, N\right)$ and $Y_i = \mathrm{Hom}_R(M_i, N)$. Let G be an arbitrary abelian group, and suppose we are given a family of group homomorphisms $\varphi_i : G \to Y_i$ for each $i \in I$. For fixed $g \in G$, the family $\{\varphi_i(g)\}_{i\in I}$ is a collection of R-module homomorphisms from M_i to N. Then by the universal property of the direct sum $\bigoplus_{i\in I} M_i$ in R-Mod, there exists a unique R-module homomorphism f_g such that

$$f_g \circ \iota_i = \varphi_i(g) \quad \text{for all } i \in I.$$

where ι_i are the canonical injections. Define $\Phi : G \to X$ by setting $\Phi(g) = f_g$. I claim that Φ is the unique group homomorphism such that the diagram

$$\begin{tikzcd} G \arrow[d, "\varphi_i"'] \arrow[dr, dashed, "\Phi"] & \\ Y_i & X \arrow[l, "\pi_i"] \end{tikzcd}$$

commutes where $\pi_i = \mathrm{Hom}_R(\iota_i, N)$ is the canonical projection induced by contravariant functor $\mathrm{Hom}_R(-, N)$. Let $g, h \in G$. Since $\varphi_i(g + h) = \varphi_i(g) + \varphi_i(h)$ we obtain that

$$f_{g+h} \circ \iota_i = \varphi_i(g) + \varphi_i(h) = (f_g + f_h) \circ \iota_i.$$

Then $f_{g+h} = f_g + f_h$, or equivalently $\Phi(g+h) = \Phi(g) + \Phi(h)$, by the uniqueness condition of the direct sum's universal property. Therefore, Φ is a group homomorphism. To prove the uniqueness of Φ, we suppose there is another group homomorphism $\Psi : G \to X$ satisfying $(\Psi(g)) \circ \iota_i = \varphi_i(g)$ for all $i \in I$ and all $g \in G$. For any fixed g, both R-module homomorphisms $\Psi(g)$ and $\Phi(g)$ pull back along ι_i to the same maps $\varphi_i(g)$. By the universal property of the direct sum, we must have $\Psi(g) = \Phi(g)$ for all g, so $\Psi = \Phi$. The uniqueness and existence of Φ yields $\mathrm{Hom}_R\left(\bigoplus_{i\in I} M_i, N\right) \simeq \prod_{i\in I} \mathrm{Hom}_R(M_i, N)$.

(2) Let $X = \mathrm{Hom}_R\left(M, \prod_{j\in J} N_j\right)$ and $Y_j = \mathrm{Hom}_R(M, N_j)$. Let G be an arbitrary abelian group, and suppose we are given a family of group homomorphisms $\varphi_j : G \to Y_j$ for each $j \in J$. Then for fixed $g \in G$ the family of R-module homomorphisms $\{\varphi_j(g) : M \to N_j\}_{j\in J}$ uniquely induced an R-module homomorphism $f_g : M \to \prod_{j\in J} N_j$ such that $\varphi_j(g) = p_j \circ f_g$. Define the map $\Phi : G \to \mathrm{Hom}_R\left(M, \prod_{j\in J} N_j\right)$ by setting $\Phi(g) = f_g$. I claim Φ is the unique group homomorphism such that the diagram

$$\begin{tikzcd} G \arrow[d, "\varphi_j"'] \arrow[dr, dashed, "\Phi"] & \\ Y_j & X \arrow[l, "\pi_j"] \end{tikzcd}$$

commutes where $\pi_j = \mathrm{Hom}_R(M, p_j)$ is the canonical projection induced by covariant functor $\mathrm{Hom}_R(M, -)$. Let $g, h \in G$. Since $\varphi_j(g + h) = \varphi_j(g) + \varphi_j(h)$ we obtain that

$$p_j \circ f_{g+h} = \varphi_j(g) + \varphi_j(h) = p_j \circ (f_g + f_h).$$

It implies $\Phi(g + h) = \Phi(g) + \Phi(h)$ because of the uniqueness condition of the direct product's universal property. Therefore, Φ is a group homomorphism. To prove the uniqueness of Φ, we suppose there is another group homomorphism $\Psi : G \to X$ satisfying $p_j \circ (\Psi(g)) = \varphi_j(g)$ for all $j \in J$ and all $g \in G$. For any fixed g, both R-module homomorphisms $\Psi(g)$ and $\Phi(g)$ push forward along p_j to the same maps $\varphi_j(g)$. By the universal property of the direct product, we must have $\Psi(g) = \Phi(g)$ for all g, so $\Psi = \Phi$. The uniqueness and existence of Φ yields $\mathrm{Hom}_R\left(M, \prod_{j\in J} N_j\right) \simeq \prod_{j\in J} \mathrm{Hom}_R(M, N_j)$. ■

Remark. The isomorphism established in **(1)** is the direct consequence of the **Hom-Hom adjunction** and the fact that right adjoints preserve limits. Specifically, the contravariant functor $\mathrm{Hom}_R(-, N)$ is the right adjoint of contravariant functor $\mathrm{Hom}_{\mathbf{Ab}}(-, N)$. More explicitly, there is a natural isomorphism

$$\mathrm{Hom}_R(M, \mathrm{Hom}_{\mathbf{Ab}}(A, N)) \cong \mathrm{Hom}_{\mathbf{Ab}}(A, \mathrm{Hom}_R(M, N)),$$

which holds for both left and right R-module categories. Similarily, **(2)** is the direct consequence of the **Tensor-Hom adjunction**. Specifically, the covariant functor $\mathrm{Hom}_R(M, -)$ is the right adjoint of the covariant functor $-\otimes_{\mathbb{Z}} M$ if we consider right R-module category and $M \otimes_{\mathbb{Z}} -$ for the left.

Problem 2

(*Five Lemma*) Suppose that the following diagram of homomorphisms of R-modules

$$\begin{tikzcd} M_1 \arrow[r, "f_1"] \arrow[d, "\varphi_1"] & M_2 \arrow[r, "f_2"] \arrow[d, "\varphi_2"] & M_3 \arrow[r, "f_3"] \arrow[d, "\varphi_3"] & M_4 \arrow[r, "f_4"] \arrow[d, "\varphi_4"] & M_5 \arrow[d, "\varphi_5"] \\ N_1 \arrow[r, "g_1"] & N_2 \arrow[r, "g_2"] & N_3 \arrow[r, "g_3"] & N_4 \arrow[r, "g_4"] & N_5 \end{tikzcd}$$

is commutative and has exact rows. Prove that

(1) if φ_1 is surjective, and φ_2 and φ_4 are injective, then φ_3 is injective;

(2) if φ_5 is injective, and φ_2 and φ_4 are surjective, then φ_3 is surjective.

Solution: **(1)** Suppose $a_3 \in M_3$ such that $\varphi_3(a_3) = 0$, then $\varphi_4 \circ f_3(a_3) = g_3 \circ \varphi_3(a_3) = 0$ implies $f_3(a_3) = 0$ since φ_4 is injective. Therefore, $\mathrm{Im}(f_2) = \ker(f_3)$ implies that there exists an element $a_2 \in M_2$ such that $f_2(a_2) = a_3$. Let $b_2 = \varphi_2(a_2)$, then $g_2(b_2) = \varphi_3 \circ f_2(a_2) = \varphi_3(a_3) = 0$ implies there exists $b_1 \in N_1$ such that $g_1(b_1) = b_2$. Because φ_1 is surjective then there exists a_1 such that $\varphi_1(a_1) = b_1$. I claim that $a_2 = f_1(a_1)$ and therefore $a_2 \in \mathrm{Im}(f_1) = \ker(f_2)$ implies $a_3 = f_2(a_2) = 0$ and thus φ_3 is injective. To prove it, we can consider $\varphi_2(a_2 - f_1(a_1)) = \varphi_2(a_2) - g_1 \circ \varphi_1(a_1) = b_2 - b_2 = 0$ which implies $a_2 - f_1(a_1) = 0$ because of injectivity of φ_2.

(2) Suppose b_3 be arbitrary elements in N_3 respectively. Let $b_4 = g_3(b_3)$. Because φ_4 is surjective then we can let $a_4 \in M_4$ such that $\varphi_4(a_4) = b_4$. Then $\varphi_5 \circ f_4(a_4) = g_4 \circ \varphi_4(a_4) = g_4 \circ g_3(b_3) = 0$ implies $f_4(a_4) = 0$ because of injectivity of φ_5. Then we can suppose $a_3 \in M_3$ such that $f_3(a_3) = a_4$. Then $g_3(b_3 - \varphi_3(a_3)) = b_4 - \varphi_4(a_4) = 0$ implies there exists $b_2 \in N_2$ such that $g_2(b_2) = b_3 - \varphi_3(a_3)$. Then the surjectivity of φ_2 implies there exists a_2 such that $\varphi_2(a_2) = b_2$. Since diagram commutes, we obtain that $\varphi_3 \circ f_2(a_2) = b_3 - \varphi_3(a_3)$, or equivalently, $b_3 = \varphi_3(f_2(a_2) + a_3) \in \mathrm{Im}(\varphi_3)$ which implies φ_3 is surjective. ■

Problem 3

(*Snake Lemma*) Suppose that the following diagram of homomorphisms of R-modules

$$\begin{tikzcd}
0 \arrow[r] & M' \arrow[r, "\alpha"] \arrow[d, "\varphi'"] & M \arrow[r, "\beta"] \arrow[d, "\varphi"] & M'' \arrow[r] \arrow[d, "\varphi''"] & 0 \\
0 \arrow[r] & N' \arrow[r, "\mu"] & N \arrow[r, "\nu"] & N'' \arrow[r] & 0
\end{tikzcd}$$

is commutative and has exact rows. Prove that

$$0 \to \ker \varphi' \xrightarrow{\alpha'} \ker \varphi \xrightarrow{\beta'} \ker \varphi'' \xrightarrow{\delta} \operatorname{coker} \varphi' \xrightarrow{\mu'} \operatorname{coker} \varphi \xrightarrow{\nu'} \operatorname{coker} \varphi'' \to 0$$

is a long exact sequence, where $\alpha', \beta', \mu', \nu'$ are homomorphisms induced by α, β, μ, ν, respectively. (Hint: for any $x \in \ker \varphi''$, take $y \in M$ such that $\beta(y) = x$, and take $z \in N'$ such that $\mu(z) = \varphi(y)$. Define $\delta(x) = \overline{z} = z + \operatorname{im} \varphi'$)

Solution: **$\boldsymbol{\alpha'}$ is well-defined and injective**. Suppose $m' \in \ker(\varphi')$, since diagram commutes, we obtain that $\varphi \circ \alpha(m') = \mu \circ \varphi'(m') = 0$ which implies $\alpha(m') \in \ker(\varphi)$. Therefore, $\alpha' = \alpha|_{\ker(\varphi')} : \ker(\varphi') \to \ker(\varphi)$ is well-defined. Hence, α' is injective since α is injective. **$\boldsymbol{\beta'}$ is well-defined and $\mathbf{Im}(\boldsymbol{\alpha'}) = \mathbf{ker}(\boldsymbol{\beta'})$.** Let $m \in \ker(\varphi)$, then $\varphi'' \circ \beta(m) = \nu \circ \varphi(m) = 0$ yields well-definedness. Suppose $m' \in \ker(\varphi')$ then $\beta' \circ \alpha'(m') = \beta \circ \alpha(m') = 0$ where $\beta \circ \alpha = 0$ because of the exactness of the first row. Therefore, we obtain that $\mathrm{Im}(\alpha') \subseteq \ker(\beta')$. Conversely, if $m \in \ker(\beta')$, then $\beta(m) = \beta'(m) = 0$. By exactness, there exists m' such that $\alpha(m') = m$. Hence, $m \in \ker(\varphi)$ implies that $\mu \circ \varphi'(m') = \varphi \circ \alpha(m') = \varphi(m) = 0$. Because μ is injective, then we obtain that $m' \in \ker(\varphi')$. Therefore, for an arbitrary element $m \in \ker(\beta')$ there exists $m' \in \ker(\varphi')$ such that $m = \alpha'(m')$ meaning that $\ker(\beta') \subseteq \mathrm{Im}(\alpha')$. **$\boldsymbol{\delta}$ is well-defined.** Because β is surjective then for any $x \in \ker(\varphi'')$ there exists an element $y \in M$ such that $\beta(y) = x$. Since $x \in \ker(\varphi'')$, we obtain that $\nu \circ \varphi(y) = \varphi''(x) = 0$, which implies $\varphi(y) \in \ker(\nu)$. By exactness, there exists an element $z \in N'$ such that $\mu(z) = \varphi(y)$. Let $y_0, y_1 \in M$ be

arbitrary two elements such that $\beta(y_0) = \beta(y_1) = x$. Let $z_0, z_1 \in N'$ be arbitrary two elements such that $\mu(z_0) = \varphi(y_0)$ and $\mu(z_1) = \varphi(y_1)$. Because $\beta(y_1 - y_0) = 0$, then there exists an element $m' \in M'$ such that $\alpha(m') = y_1 - y_0$. Therefore,

$$\begin{aligned}\mu(\varphi'(m') - z_1 + z_0) &= \mu \circ \varphi'(m') - \mu(z_1 - z_0) \\ &= \varphi \circ \alpha(m') - \mu(z_1 - z_0) \\ &= \varphi(y_1 - y_0) - \mu(z_1 - z_0) = 0\end{aligned}$$

implies $\varphi'(m') = z_1 - z_0$ since μ is injective. Equivalently, we obtain that $\overline{z_1 - z_0} = 0$ which yields the well-definedness of δ. $\mathbf{Im(\beta') = ker(\delta)}$. Suppose $m \in \ker(\varphi)$, then we can select $z = 0 \in N'$ because $\mu(z) = \varphi(m) = 0$ which implies $\delta \circ \beta'(m) = 0$. Therefore, $\mathrm{Im}(\beta') \subseteq \ker(\delta)$. Suppose $m'' \in \ker(\delta)$, then there exists $m \in M$ and $n' \in N'$ such that $\beta(m) = m''$ and $\varphi(m) = \mu(n')$. Hence $\delta(m'') = 0$ implies there exists an element $m' \in M'$ such that $\varphi'(m') = n'$. I claim that $m - \alpha(m') \in \ker(\varphi)$ and thus $\beta'(m - \alpha(m')) = \beta(m) - \beta \circ \alpha(m') = m''$ implies that $\ker(\delta) \subseteq \mathrm{Im}(\beta')$. To see that, we consider $\varphi(m - \alpha(m')) = \varphi(m) - \mu \circ \varphi'(m') = \varphi(m) - \mu(n') = 0$. $\boldsymbol{\mu'}$ **is well-defined and** $\mathbf{Im(\delta) = ker(\mu')}$**.** Since the diagram commutes, $\mu \circ \varphi' = \varphi \circ \alpha$ implies $\mu(\mathrm{Im}(\varphi')) \subseteq \mathrm{Im}(\varphi)$ which yields well-definedness. Let m'' be an arbitrary element in $\ker(\varphi'')$ then there exists $m \in M$ and $n' \in N'$ such that $\beta(m) = m''$ and $\varphi(m) = \mu(n')$. Then we have $\mu' \circ \delta(m'') = \mu'(n' + \mathrm{Im}(\varphi')) = \mu(n') + \mathrm{Im}(\varphi) = 0 \in \mathrm{coker}(\varphi)$, which implies $\mathrm{Im}(\delta) \subseteq \ker(\mu')$. For an arbitrary element $\overline{n'} \in \ker(\mu')$, then we have $\mu(n') \in \mathrm{Im}(\varphi)$, which implies there exists an element $m \in M$ such that $\varphi(m) = \mu(n')$. Let $m'' = \beta(m)$ and I claim that $m'' \in \ker(\varphi'')$ and thus $\delta(m'') = \overline{n'}$ implies $\ker(\mu') \subseteq \mathrm{Im}(\delta)$. To see that, just consider $\varphi''(m'') = \nu \circ \varphi(m) = \nu \circ \mu(n') = 0$. $\boldsymbol{\nu'}$ **is well-defined and** $\mathbf{ker(\nu') = Im(\mu')}$**.** Similarily, because diagram commutes, then $\nu \circ \varphi = \varphi'' \circ \beta$ implies that $\nu(\mathrm{Im}(\varphi)) \subseteq \mathrm{Im}(\varphi'')$ meaning that ν' is well-defined. Let $\overline{n'}$ be an arbitrary element in $\mathrm{coker}(\varphi')$, then $\nu' \circ \mu'(\overline{n'}) = \nu'(\mu(n') + \mathrm{Im}(\varphi)) = \nu \circ \mu(n') + \mathrm{Im}(\varphi'') = 0 \in \mathrm{coker}(\varphi'')$. Therefore, $\mathrm{Im}(\mu') \subseteq \ker(\nu')$. Let $\overline{n}$ be an arbitrary element such that $\nu'(\overline{n}) = 0$, equivalently $\nu(n) \in \mathrm{Im}(\varphi'')$. Then there exists an element $m'' \in M''$ such that $\varphi''(m'') = \nu(n)$. Since β is surjective, then there exists an element $m \in M$ such that $\beta(m) = m''$. I claim that $n - \varphi(m) \in \ker(\nu)$ and thus there exists an element $n' \in N'$ such that $\mu(n') = n - \varphi(m)$, which implies that $\mu'(\overline{n'}) = \overline{n}$ meaning that $\ker(\nu') \subseteq \mathrm{Im}(\mu')$. To see that, we consider $\nu(n - \varphi(m)) = \nu(n) - \varphi'' \circ \beta(m) = \nu(n) - \varphi''(m) = 0$. $\boldsymbol{\nu'}$ **is surjective.** It is directly because ν' is well-defined and induced by surjection ν. Explicitly, for any arbitrary element $\overline{n''} \in \mathrm{coker}(\varphi'')$, there exists an element $n \in N$ such that $\nu(n) = n''$. Since $\nu(\mathrm{Im}(\varphi)) \subseteq \mathrm{Im}(\varphi'')$ we have discussed, then we have $\nu'(\overline{n}) = n'' + \mathrm{Im}(\varphi'') = \overline{n''}$. ■

Lemmas for Problem 3

Lemma A. *Let $f : M \to M''$ and $g : K \to M''$ be R-module homomorphisms. Let $P = M \times_{M''} K$ be the pullback of f along g, defined explicitly by the subset of the direct sum:*

$$P = \{(m, k) \in M \times K \mid f(m) = g(k)\}$$

equipped with the canonical projections $\pi_M : P \to M, (m, k) \mapsto m$ and $\pi_K : P \to K, (m, k) \mapsto k$. Then the following properties hold:

1. *If f is surjective, then the projection π_K is surjective.*
2. *If f is injective, then the projection π_K is injective.*

Proof. **(1)** Suppose f is surjective. Let $k \in K$ be an arbitrary element. Because f is surjective, then there exists an element $m \in M$ such that $f(m) = g(k)$, which implies $(m, k) \in P$ then $k = \pi_K(m, k)$ implies surjectivity.

(2) Suppose f is injective. Suppose $(m, k) \in P$ such that $\pi_K(m, k) = 0$. Then we obtain that $f \circ \pi_M(m, k) = f(m) = g(k) = g \circ \pi_K(m, k) = 0$. The injectivity of f implies $m = 0$. Hence, $k = \pi_K(m, k) = 0$ implies the injectivity of π_K. □

Lemma B. *Let $\mu : N' \to N$ and $h : N' \to L$ be R-module homomorphisms. Let $Q = N \oplus_{N'} L$ be the pushout of μ along h, defined explicitly as the quotient module of the direct sum:*

$$Q = (N \oplus L)/W$$

where $W = \{(\mu(n'), -h(n')) \mid n' \in N'\}$. The induced maps are given by $\bar{h} : N \to Q, n \mapsto \overline{(n, 0)}$ and $i_L : L \to Q, l \mapsto \overline{(0, l)}$. Then the following properties hold:

1. *If μ is surjective, then the induced map i_L is surjective.*
2. *If μ is injective, then the induced map i_L is injective.*

Proof. **(1)** Suppose μ is surjective. It suffices to prove that any pair with form $\overline{(n, 0)}$ has preimage because $\overline{(n, l)} = \overline{(n, 0)} + \overline{(0, l)}$ and $\overline{(0, l)}$ has preimage. Because μ is surjective, then there exists $n' \in N'$ such that $\mu(n') = n$. I claim that $\overline{(n, 0)} = \overline{(0, h(n'))} = i_L(h(n'))$ and thus surjectivity of i_L holds. To see that, we consider $\overline{(n, 0)} - \overline{(0, h(n'))} = \overline{(n, -h(n'))} = 0$ since $n = \mu(n')$.

(2) Suppose μ is injective. Let $l \in L$ be an arbitrary element such that $i_L(l) = \overline{(0, l)} = 0$, or equivalently there exists $n' \in N'$ such that $\mu(n') = 0$ and $-h(n') = l$. The injectivity of μ implies $n' = 0$ and hence $l = -h(n') = 0$. Therefore, i_L is injective. □